

Product Analytics

Time to Conversion

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Introduction to the project

This analysis focuses on understanding how long it takes users to complete their first purchase after arriving on the website each day. By measuring the duration between a user's first visit and their first purchase on the same day, we can better understand customer behavior, conversion efficiency, and purchasing patterns over time.

Using data from the `turing_data_analytics.raw_events` table, we analyzed the daily progression of purchase duration and explored additional behavioral insights across users and sessions. The goal of this analysis is to identify trends, highlight potential opportunities for improving the conversion journey, and support data-driven decisions for marketing and product optimization.

In this presentation, we will explain the methodology, visualize key findings, discuss limitations of the analysis, and recommend further areas for investigation.



Key Questions & Business Value

Key Questions

- How do purchase times differ across different periods?
- How do purchase times differ across weekdays?
- How do purchase times vary during different parts of the day?
- How do purchase times differ by country?
- Does device type impact the time users take to purchase?

Business Value

This analysis helps the business:

- Identify high-converting customer behaviors
- Improve campaign timing and targeting
- Optimize regional strategies
- Enhance customer experience
- Support data-driven decision making



Data Exploration & Preparation

- Explored the `raw_events` dataset to understand available customer interaction, purchase, geographical, and device data.
- Identified the key fields required to answer the business questions and support the analysis.
- Created a SQL query in BigQuery to extract the relevant data for visualization and reporting.
- Developed additional calculated fields, including:
 - **Purchase Time of Day** (Morning, Afternoon, Evening, Night)
 - **Time Difference** – duration in minutes between the user's first page view and first purchase on the same day.
- Prepared the final dataset for analysis and Tableau dashboard creation.

Key Business Metrics



Total Customers

4,419

Total Orders

5,691

Revenue

\$362,165.00

Avg. Revenue

\$63.64

Avg. Duration in Minu..

79 Minutes

Unique Customers: 4,419 customers made at least one purchase.

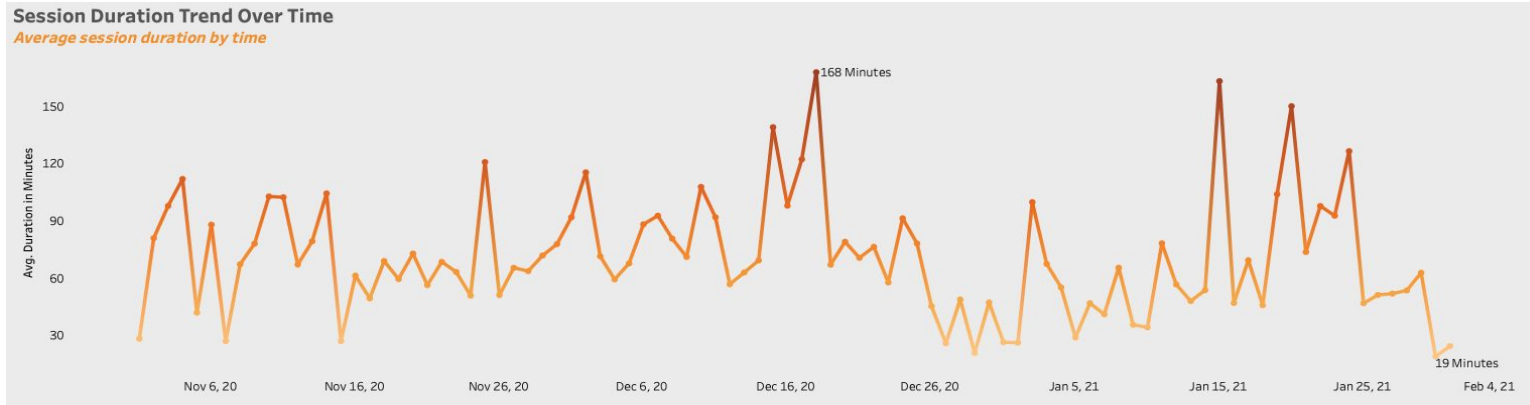
Total Orders: Users completed 5,691 purchases.

Total Revenue: Purchases generated approximately **\$362K** in revenue.

Average Order Value: The average revenue per order was **\$64**.

Average Time to Purchase: Users took an average of **79 minutes** from their first page view to completing a purchase on the same day.

How Purchase Behavior Changes over Time



- The chart shows daily average time users spent before completing their first purchase from Nov 2020 to Feb 2021.
- Purchase duration fluctuated throughout the period, showing changes in customer behavior over time.
- The highest average duration was recorded in mid-December at approximately **168 minutes**.
- Longer browsing times during December may indicate users were comparing products and searching for holiday gifts.
- Another increase in session duration appeared during mid-to-late January, possibly influenced by post-holiday promotions and discounts.
- By early February, the average purchase duration dropped significantly to around **19 minutes**.

Top Performing Countries by Revenue & Purchase Duration

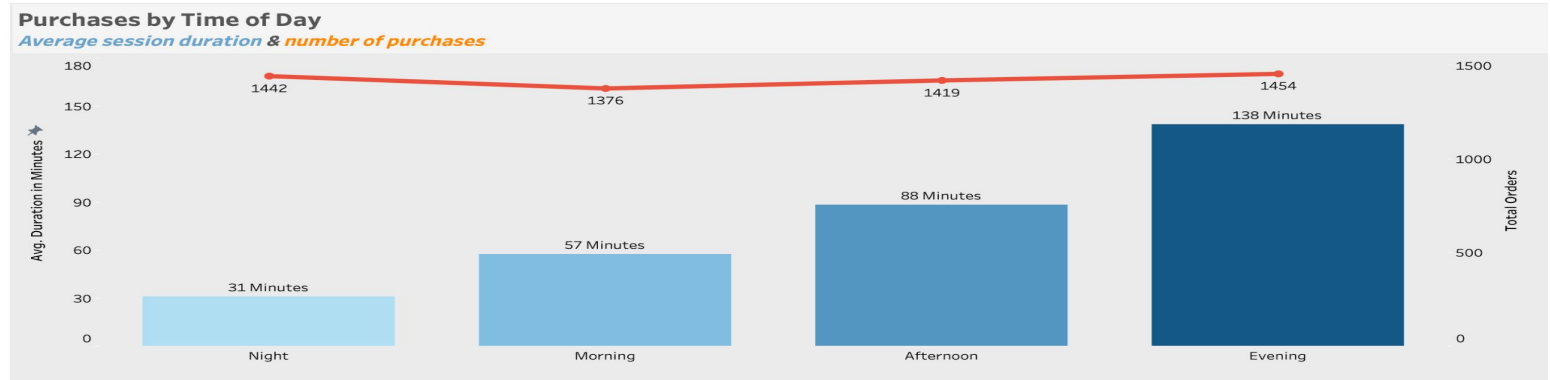
Top 10 Countries

Revenue & Average Session Time

United States	\$160,573.00	76 Minutes
India	\$34,986.00	65 Minutes
Canada	\$32,799.00	65 Minutes
United Kingdom	\$11,458.00	95 Minutes
Spain	\$7,681.00	106 Minutes
France	\$6,650.00	97 Minutes
China	\$6,623.00	87 Minutes
Japan	\$5,752.00	54 Minutes
Turkey	\$5,345.00	107 Minutes
Germany	\$5,288.00	59 Minutes

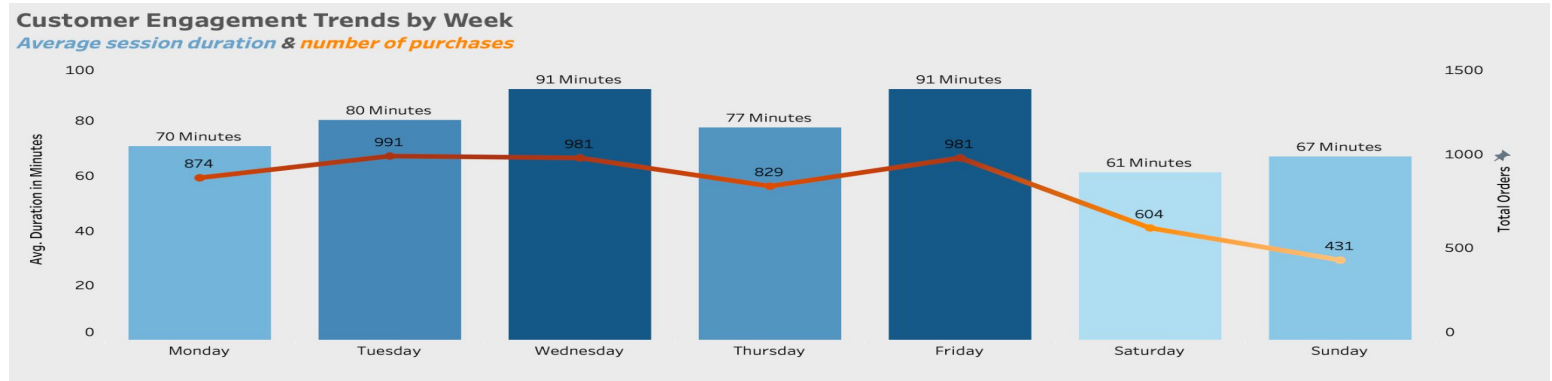
- The United States generated the highest revenue by a significant margin, with over **\$160K** in total sales.
- India and Canada were the second and third highest revenue-generating countries, both with relatively lower average purchase durations of around **65 minutes**.
- Countries such as Spain, France, and Turkey showed longer average purchase durations, exceeding **95–107 minutes**.
- Longer session durations may indicate that customers in these regions spend more time comparing products before purchasing.
- Japan and Germany had some of the shortest average purchase durations, suggesting faster purchasing decisions and potentially more efficient customer journeys.
- The analysis highlights clear differences in purchasing behavior across countries, emphasizing the importance of regional marketing and localization strategies.

Purchases by Time of Day



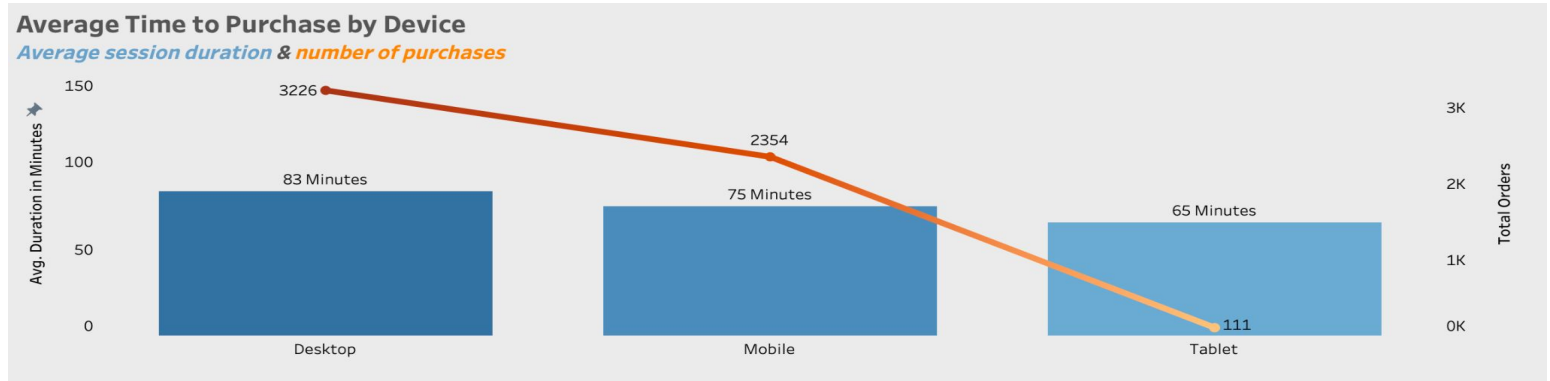
- The chart compares the average purchase duration and total number of purchases across different parts of the day.
- Evening recorded the highest average purchase duration at approximately **138 minutes**.
- Night had the shortest average purchase duration at around **31 minutes**, suggesting faster purchasing decisions during late hours.
- Purchase duration gradually increased throughout the day from morning to evening.
- Despite differences in browsing time, the total number of purchases remained relatively stable across all time periods, ranging between **1,376 and 1,454 orders**.
- The longer browsing times during the evening may indicate that users spend more time comparing products and exploring options after working hours.

Customer Engagement Trends by Weekday



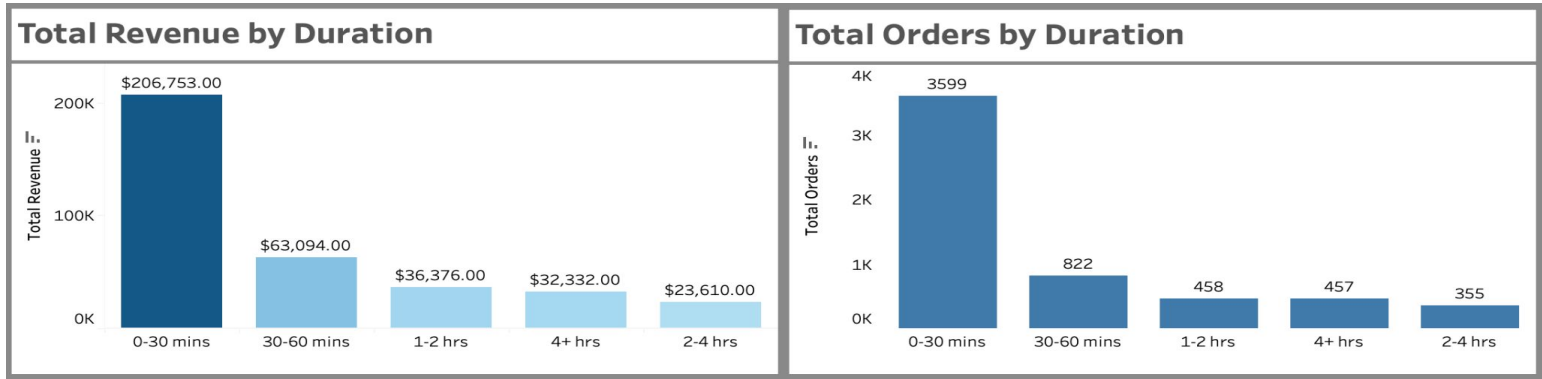
- The chart compares average purchase duration and total number of purchases across weekdays.
- Wednesday and Friday recorded the highest average purchase duration at approximately **91 minutes**.
- Tuesday generated the highest number of purchases with **991 orders**.
- Purchase activity remained relatively strong during weekdays but declined significantly over the weekend.
- Saturday and Sunday showed the lowest number of purchases, with only **604** and **431** orders respectively.
- Weekend users also spent less time browsing before purchasing, which may indicate lower engagement levels compared to weekdays.
- The analysis suggests that weekdays, particularly mid-week, are the strongest periods for customer engagement and purchasing activity.

Average Time to Purchase by Device



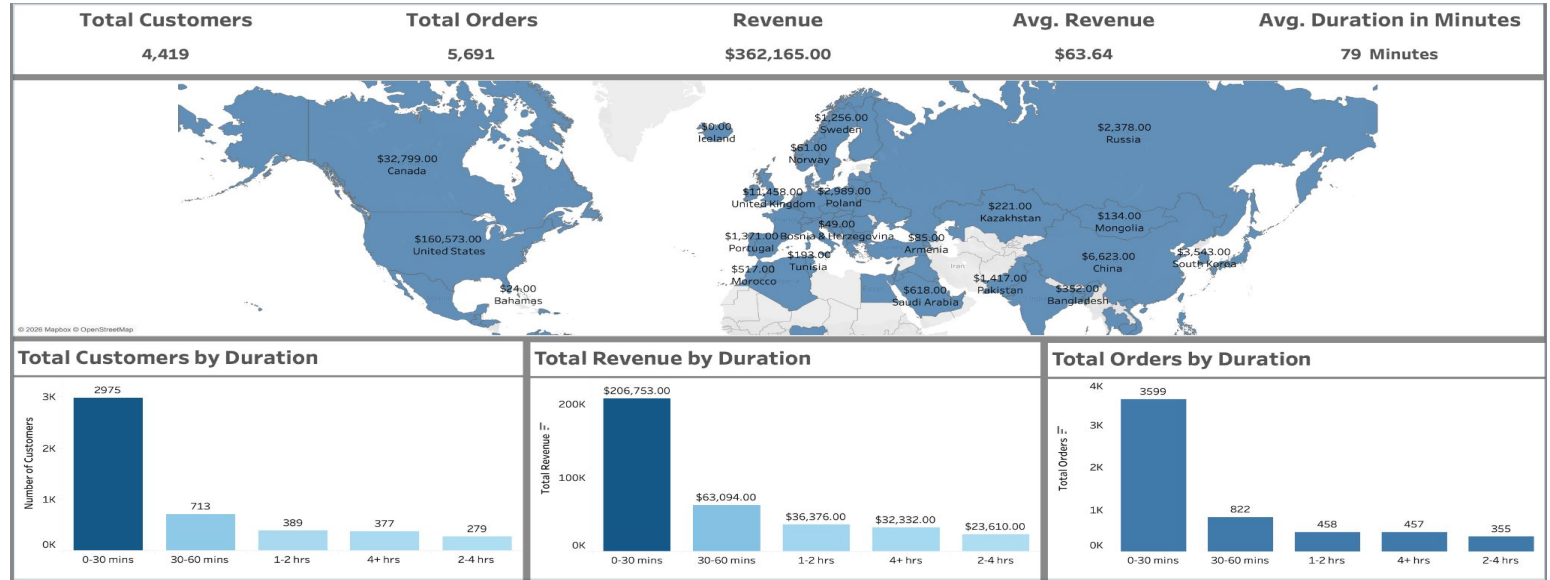
- The chart compares average purchase duration and total number of purchases across device types.
- Desktop users generated the highest number of purchases with **3,226 orders** and also had the longest average purchase duration at **83 minutes**.
- Mobile devices accounted for **2,354 purchases** with an average purchase duration of **75 minutes**.
- Tablet users had the lowest engagement, generating only **111 orders** and the shortest average purchase duration at **65 minutes**.
- The results suggest that desktop users spend more time browsing and interacting with products before completing a purchase.
- Mobile devices also play a significant role in customer purchases, highlighting the importance of mobile-friendly user experience and optimization.

Revenue & Orders by Purchase Duration



- The majority of purchases were completed within the first **30 minutes** after users visited the website.
- The **0–30 minute** duration group generated the highest revenue, contributing over **\$206K** and **3,599 total orders**.
- As purchase duration increased, both revenue and number of orders gradually decreased.
- Users who completed purchases quickly were the most valuable customer segment in terms of overall revenue generation.
- Longer purchase durations, such as **2–4 hours** and **4+ hours**, generated significantly fewer orders and lower revenue.
- The analysis suggests that faster customer conversions are strongly associated with higher purchasing activity and revenue performance.

Purchase Duration by Countries



This interactive dashboard provides a deeper regional analysis of customer purchasing behavior across different countries. By selecting a country directly from the map, all KPIs and visualizations dynamically update to display country-specific insights, including total customers, orders, revenue, and purchase duration distribution.

Final Conclusions



- The analysis revealed that most users convert relatively quickly, with an average purchase duration of **79 minutes** and an average order value of approximately **\$64**.
- Customer behavior varies significantly depending on time period, weekday, country, and device type, demonstrating that purchasing behavior is influenced by multiple external and behavioral factors.
- Seasonal trends showed increased browsing activity during the pre-Christmas and post-holiday periods, suggesting that promotional campaigns and holiday shopping strongly affect customer decision-making time.
- Evening hours and mid-week days generated the highest engagement levels, indicating that customers are more active and spend more time exploring products during these periods.
- Regional analysis highlighted major differences in customer behavior across countries. The United States was the strongest-performing market in terms of revenue and customer activity, while other regions showed longer purchasing journeys and different engagement patterns.
- Device analysis demonstrated that desktop users generated the highest number of purchases and spent more time browsing before conversion, while mobile devices also represented a significant portion of total purchases, emphasizing the importance of cross-device optimization.
- One of the strongest findings from the analysis was that the majority of revenue and orders came from users who completed purchases within the first **30 minutes**, suggesting that faster customer conversions are closely linked to higher business performance.
- Finally, the analysis identified a smaller segment of users with unusually long purchase durations. Further investigation may help determine whether these sessions represent outliers, complex customer journeys, or opportunities for improving website usability and conversion flow.

Final Recommendation

The business should focus on optimizing the early stages of the customer journey, improving mobile and desktop user experience, and aligning marketing campaigns with peak engagement periods. Additional future analysis could include customer segmentation, product category analysis, traffic source performance, and funnel analysis to better understand conversion behavior and further improve customer experience.